

A change in the value of a parameter leads to a shift in the IS curve. An increase in autonomous consumption (the coefficient 'a' in equation (3.4)) or autonomous investment (the coefficient 'c' in equation (3.4)) will lead to a parallel shift of the IS curve upward to the right. A change in the marginal propensity to consume, however, will affect the slope of the IS curve.

Check Your Progress 1

- 1) What will be the nature of change in the IS curve if there is a decrease in the propensity to consume?

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- 2) What is the implication of the following on the position of the IS curve?

(i) Decrease in autonomous government expenditure

(ii) Decrease in tax rate

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3.3 EQUILIBRIUM IN THE MONETARY SECTOR

The derivation of LM curve utilises the Keynesian view that speculative demand and transaction demand for money are separate. The speculative demand for money depends on the rate of interest while the transaction demand and the precautionary demand for money depend on the level of income. Thus, total demand for money (M) depends on both income (Y) and rate of interest (i). The demand for money in real terms ($\frac{M^d}{P}$) is given by

$$L = kY - hi \quad \dots (3.5)$$

where $k > 0$, $h > 0$; L is the demand for real balances (liquidity), Y is aggregate income and i is the rate of interest. You should remember three issues about the monetary sector: (i) A person would like to deposit his money in bank and earn interest on it. The alternative is to hold on to the cash, where no interest income is earned. Thus, the opportunity cost of holding money is the interest foregone. If there is an increase in the rate of interest, the interest foregone increases. Conversely, if the rate of interest decreases, the interest income foregone decreases. Therefore, the demand for money is a downward-sloping curve, if we measure rate of interest on the y-axis and quantity of money on the x-axis. You may be familiar with the concept of liquidity trap; a situation where the rate of interest is very low, and people prefer to keep all their money in the form of liquid cash as the interest income foregone is negligible. (ii) The demand for real balances is described for a given level of income. We consider the money demand function

while keeping the level of income fixed (say, $L(Y_0)$). If there is an increase in the level of income, there is an increase in the level of transactions and wealth in the economy, which in turn increases the demand for money. Thus, there will be a parallel shift in the money demand function. (iii) The supply of money is determined by the central bank and it is constant. Thus, the supply of money is a vertical straight line. If the supply of money is increased by the central bank, the vertical straight line will shift to the right. When we deal with the short-run, we assume that money supply does not change.

3.3.1 Derivation of the LM Curve

The supply of money in real terms is given by $\frac{M^s}{P}$. Equilibrium in the money market is achieved when $L = M = \bar{M}$. Thus, equilibrium in the money market is given by

$$\frac{\bar{M}}{P} = kY - hi \quad \dots (3.6)$$

We can re-arrange terms in the above equation and obtain

$$Y = -\frac{1}{h} \frac{\bar{M}}{P} + \frac{k}{h} i \quad \dots (3.6a)$$

There is a positive relationship between Y and i in the LM curve. In order to find out the slope of equation (3.6) you have to express the equation in terms i and find out the slope $\frac{di}{dY}$. You can find out the slope of the LM curve as $\frac{k}{h}$. Depending upon the value of the parameters, the LM curve would be flatter or steeper.

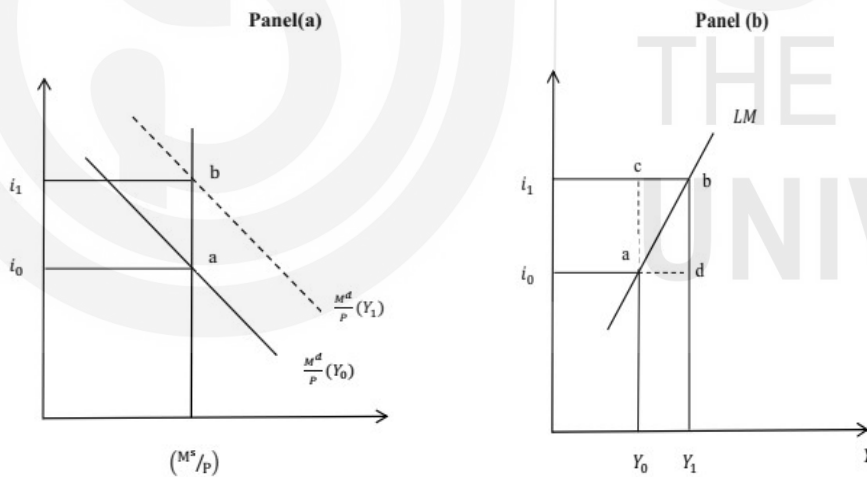


Fig. 3.3: Derivation of the LM Curve

In Fig. 3.3, Panel (a) we depict the equilibrium in the money market (or, in the monetary sector). We take demand for real balances ($\frac{M}{P}$) on the x-axis and the rate of interest on the y-axis. Let the money demand in the economy be given as $\frac{M^d}{P}(Y_0)$ when the aggregate income is Y_0 . In panel (a) of Fig. 3.3, this equilibrium is represented by point 'a'. The equilibrium rate of interest and income are i_0 and Y_0 respectively.

Let there be an increase in the level of income from Y_0 to Y_1 . Such an increase in income will lead to a parallel upward shift in the money demand function. We

depict this money demand function as a dotted line $\frac{M^d}{P}(Y_1)$ in Panel (a) of Fig. 3.3. With money supply remaining constant, an increase in aggregate demand will lead to a rise in the interest rate. The economy is now at equilibrium with output level Y_1 and interest rate i_1 . The intersection between the money supply function and the money demand function is denoted by point 'b' in Fig. 3.3, panel (a).

In panel (b) of Fig. 3.3 we plot the LM curve by taking combinations of interest rate and aggregate income where the money market is in equilibrium. We have two such points, 'a' and 'b' in panel (a) of Fig. 3.3. You should note that the LM curve is the combination of all the points where the money market is in equilibrium. The LM curve is upward-sloping as you can see from the figure. An implication of the LM curve is that as income increases people want to hold more money, thus driving up the interest rate. If the speculative demand for money is perfectly elastic to interest rate (liquidity trap situation) the LM curve will be horizontal. On the other hand, if speculative demand is perfectly inelastic to interest rate, the LM curve will be vertical.

To the left side of the LM curve there is excess supply of money (for example, point 'c' in Fig. 3.3 panel (b)). Excess supply in the money market implies that the demand for money is less than its supply. It means that people want to reduce the amount of liquid cash they possess. This induces people to buy bonds so that bond prices rise and rate of interest declines. A decline in the rate of interest restores equilibrium in the economy. To the right side of the LM curve, there is excess demand for money (for example, point 'd' in Fig. 3.3 panel (b)). Excess demand for money causes people to sell bonds (so that they can keep higher amount of liquid cash). This leads to a fall in bond prices and rise in the rate of interest. This way equilibrium in the monetary sector is restored.

3.3.2 Shift in the LM Curve

The position and the slope of the LM curve depend upon certain factors, including money supply. An increase in money supply shifts the LM curve to the right. As a result, at every level of output, the equilibrium rate of interest is lower than before. Thus, increase in money supply shifts the LM curve to the right. Conversely, a decrease in money supply will shift the LM curve to the left so that equilibrium rate of interest is higher at every level of output.

3.4 SIMULTANEOUS EQUILIBRIUM OF REAL AND MONETARY SECTORS

Now let us combine IS and LM curves as shown in Fig. 3.4. Such integration of IS-LM gives a unique combination of i and Y , which represents equilibrium in both real market and money market. In Fig. 3.4 we superimpose IS_0 and LM_0 in the same diagram. We measure income (Y) on the x-axis and rate of interest on the y-axis. The equilibrium rate of interest and the equilibrium output level are given by i_0 and Y_0 respectively.